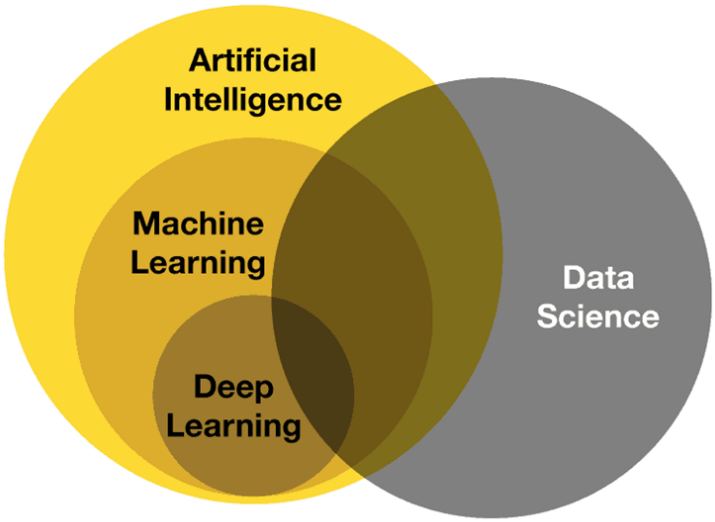
UNIT – I

Artificial Intelligence

Artificial intelligence is a field of computer science which makes a computer system that can mimic human intelligence. It is comprised of two words "Artificial" and "intelligence", which means "a human-made thinking power." Hence, we can define it as,

Artificial intelligence is a technology using which we can create intelligent systems that can simulate human intelligence.

The Artificial intelligence system does not require to be pre-programmed, instead of that, they use such algorithms which can work with their own intelligence. It involves machine learning algorithms such as Reinforcement learning algorithm and deep learning neural networks. AI is being used in multiple places such as Siri, Google, AlphaGo, AI in Chess playing, etc.



Introduction to Machine Learning

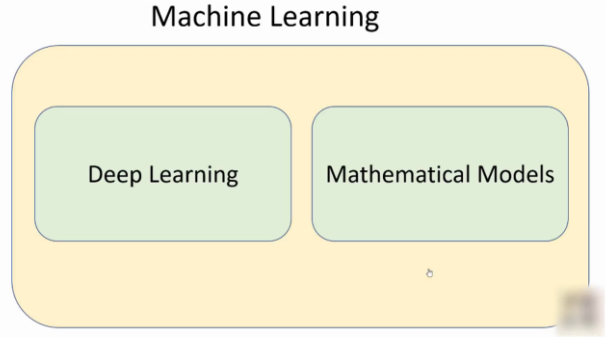
A subset of artificial intelligence known as machine learning focuses primarily on the creation of algorithms that enable a computer to independently learn from data and previous experiences. Arthur Samuel first used the term "machine learning" in 1959. It could be summarized as follows:

Without being explicitly programmed, machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things.

Machine learning algorithms create a mathematical model that, without being explicitly programmed, aids in making predictions or decisions with the assistance of sample historical data, or training data. For the purpose of

developing predictive models, machine learning brings together statistics and computer science. Algorithms that learn from historical data are either constructed or utilized in machine learning. The performance will rise in proportion to the quantity of information we provide.

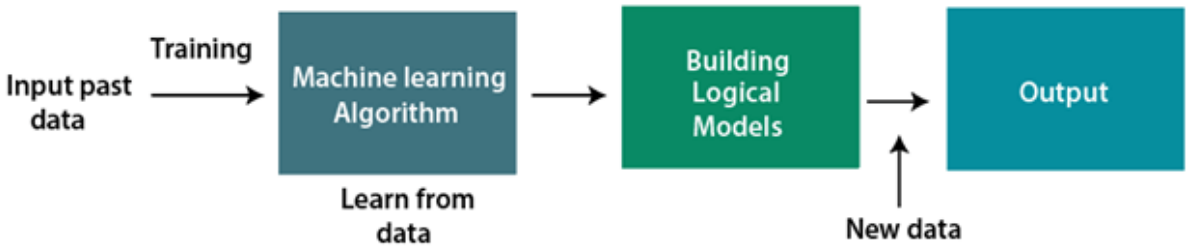
A machine can learn if it can gain more data to improve its performance.



How does Machine Learning work

A machine learning system builds prediction models, learns from previous data, and predicts the output of new data whenever it receives it. The amount of data helps to build a better model that accurately predicts the output, which in turn affects the accuracy of the predicted output.

Let's say we have a complex problem in which we need to make predictions. Instead of writing code, we just need to feed the data to generic algorithms, which build the logic based on the data and predict the output. Our perspective on the issue has changed as a result of machine learning. The Machine Learning algorithm's operation is depicted in the following block diagram:



Train Test Split Using Sklearn

The **train_test_split()** method is used to split our data into train and test sets. First, we need to divide our data into features (X) and labels (y).

The dataframe gets divided into X_train, X_test , y_train and y_test. X_train and y_train sets are used for training and fitting the model.

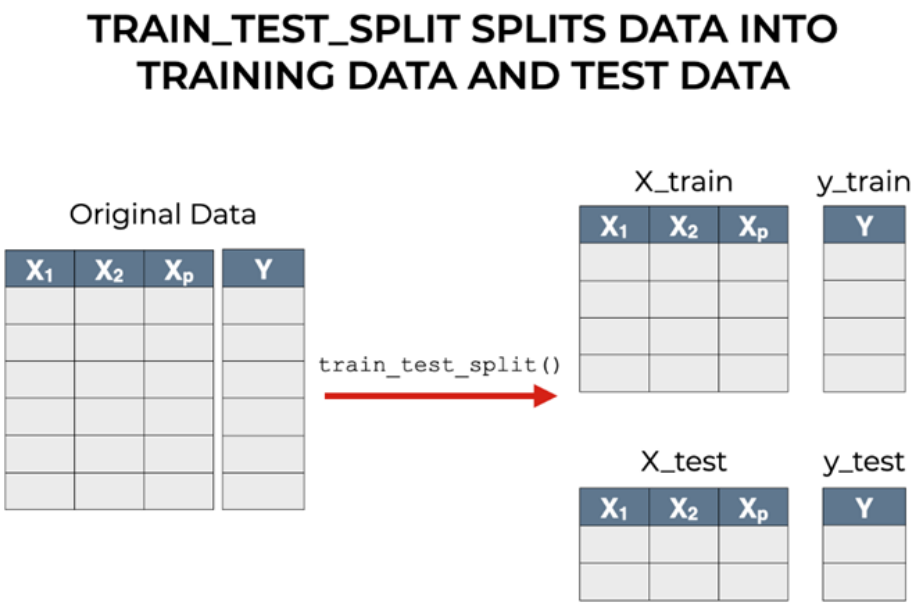
The X_test and y_test sets are used for testing the model if it's predicting the right outputs/labels. we can explicitly test the size of the train and test sets. It is suggested to keep our train sets larger than the test sets.

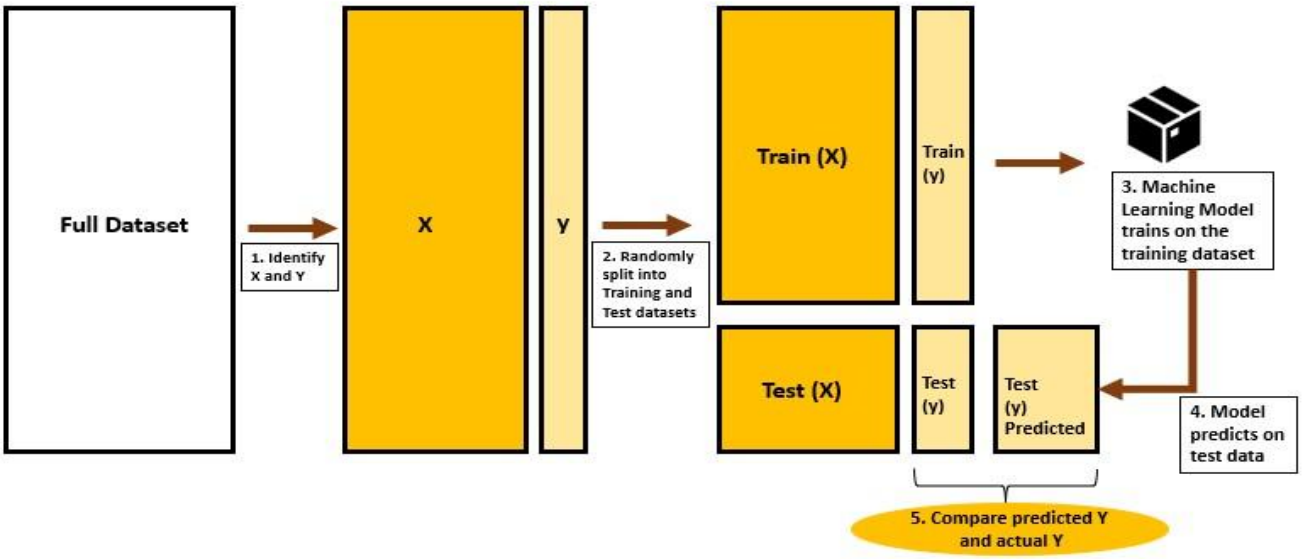
Train set: The training dataset is a set of data that was utilized to fit the model. The dataset on which the model is trained. This data is seen and learned by the model.

Test set: The test dataset is a subset of the training dataset that is utilized to give an accurate evaluation of a final model fit.

validation set: A validation dataset is a sample of data from your model's training set that is used to estimate model performance while tuning the model's hyperparameters.

	Mileage	Age(yrs)	Sell Price($)
Training (80%)	69000	6	18000
	35000	3	34000
	57000	5	26100
	22500	2	40000
	46000	4	31500
	59000	5	26750
	52000	5	32000
	72000	6	19300
Test	91000	8	12000
	67000	6	22000





Syntax:
sklearn.model_selection.train_test_split(datasets,test_size,random_state)

parameters:

arrays: Datasets

test_size: int or float, by default None.

train_size: int, by default None.

random_state: int,by default None. Controls how the data is shuffled before the split is implemented. For repeatable output across several function calls,pass an int.

shuffle: boolean object , by default True. Whether or not the data should be shuffled before splitting. Stratify must be None if shuffle=False.

stratify: array-like object, by default it is None. If None is selected, the data is stratified using these as class labels.

splitting: The train-test split of inputs is represented as a list.

Features of Machine Learning:

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.

- Machine learning is much similar to data mining as it also deals with the huge amount of the data.

Need for Machine Learning

The demand for machine learning is steadily rising. Because it is able to perform tasks that are too complex for a person to directly implement, machine learning is required. Humans are constrained by our inability to manually access vast amounts of data; as a result, we require computer systems, which is where machine learning comes in to simplify our lives.

By providing them with a large amount of data and allowing them to automatically explore the data, build models, and predict the required output, we can train machine learning algorithms. The cost function can be used to determine the amount of data and the machine learning algorithm's performance. We can save both time and money by using machine learning.

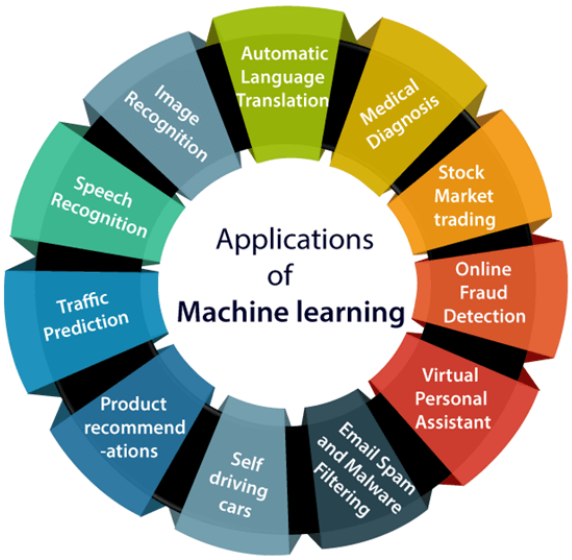
The significance of AI can be handily perceived by its utilization's cases, Presently, AI is utilized in self-driving vehicles, digital misrepresentation identification, face acknowledgment, and companion idea by Facebook, and so on. Different top organizations, for example, Netflix and Amazon have constructed AI models that are utilizing an immense measure of information to examine the client interest and suggest item likewise.

Following are some key points which show the importance of Machine Learning:

- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Decision making in various sector including finance
- Finding hidden patterns and extracting useful information from data.

Applications of Machine learning

Machine learning is a buzzword for today's technology, and it is growing very rapidly day by day. We are using machine learning in our daily life even without knowing it such as Google Maps, Google assistant, Alexa, etc. Below are some most trending real-world applications of Machine Learning:



1. Image Recognition:

Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, **Automatic friend tagging suggestion**:

Facebook provides us a feature of auto friend tagging suggestion. Whenever we upload a photo with our Facebook friends, then we automatically get a tagging suggestion with name, and the technology behind this is machine learning's **face detection** and **recognition algorithm**.

It is based on the Facebook project named "**Deep Face**," which is responsible for face recognition and person identification in the picture.

2. Speech Recognition

While using Google, we get an option of "**Search by voice**," it comes under speech recognition, and it's a popular application of machine learning.

Speech recognition is a process of converting voice instructions into text, and it is also known as "**Speech to text**", or "**Computer speech recognition**." At present, machine learning algorithms are widely used by various applications of speech recognition. **Google assistant**, **Siri**, **Cortana**, and **Alexa** are using speech recognition technology to follow the voice instructions.

3. Traffic prediction:

If we want to visit a new place, we take help of Google Maps, which shows us the correct path with the shortest route and predicts the traffic conditions.

It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:

- **Real Time location** of the vehicle form Google Map app and sensors
- **Average time has taken** on past days at the same time.

Everyone who is using Google Map is helping this app to make it better. It takes information from the user and sends back to its database to improve the performance.

4. Product recommendations:

Machine learning is widely used by various e-commerce and entertainment companies such as **Amazon**, **Netflix**, etc., for product recommendation to the user. Whenever we search for some product on Amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.

Google understands the user interest using various machine learning algorithms and suggests the product as per customer interest.

As similar, when we use Netflix, we find some recommendations for entertainment series, movies, etc., and this is also done with the help of machine learning.

5. Self-driving cars:

One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars. Tesla, the most popular car manufacturing company is working on self-driving car. It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering:

Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning.

Some machine learning algorithms such as **Multi-Layer Perceptron**, **Decision tree**, and **Naïve Bayes classifier** are used for email spam filtering and malware detection.

7. Virtual Personal Assistant:

We have various virtual personal assistants such as **Google assistant**, **Alexa**, **Cortana**, **Siri**. As the name suggests, they help us in finding the information using our voice instruction. These assistants can help us in various ways just by our voice instructions such as Play music, call someone, open an email, Scheduling an appointment, etc.

These virtual assistants use machine learning algorithms as an important part.

These assistants record our voice instructions, send it over the server on a cloud, and decode it using ML algorithms and act accordingly.

8. Online Fraud Detection:

Machine learning is making our online transaction safe and secure by detecting fraud transaction. Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place such as **fake accounts**, **fake ids**, and **steal money** in the middle of a transaction. So, to detect this, **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud transaction.

For each genuine transaction, the output is converted into some hash values, and these values become the input for the next round. For each genuine transaction, there is a specific pattern which gets change for the fraud transaction hence, it detects it and makes our online transactions more secure.

9. Stock Market trading:

Machine learning is widely used in stock market trading. In the stock market, there is always a risk of up and downs in shares, so for this machine learning's **long short term memory neural network** is used for the prediction of stock market trends.

10. Medical Diagnosis:

In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact position of lesions in the brain.

It helps in finding brain tumours and other brain-related diseases easily.

11. Automatic Language Translation:

Nowadays, if we visit a new place and we are not aware of the language then it is not a problem at all, as for this also machine learning helps us by converting the text into our known languages. Google's GNMT (Google Neural Machine Translation) provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

The technology behind the automatic translation is a sequence-to-sequence learning algorithm, which is used with image recognition and translates the text from one language to another language.

Learning in Machine Learning:

Learning in Machine Learning is the process by which a computer system **automatically improves its performance** on a task by **learning patterns from data**, rather than being explicitly programmed for every rule.

A classic way to understand ML learning is through **Tom Mitchell's definition**:

*A computer program is said to learn from experience **E**, with respect to some task **T** and performance measure **P**, if its performance at task **T**, as measured by **P**, improves with experience **E**.*

What Does "Learning" Mean in ML?

In ML, learning means:

- Adjusting **model parameters** (weights, thresholds)
- Minimizing **error or loss**
- Improving **predictions or decisions** over time

Example:

- Email spam filter learns from labelled emails
- Face recognition improves as more images are provided
- Stock price model learns trends from historical data

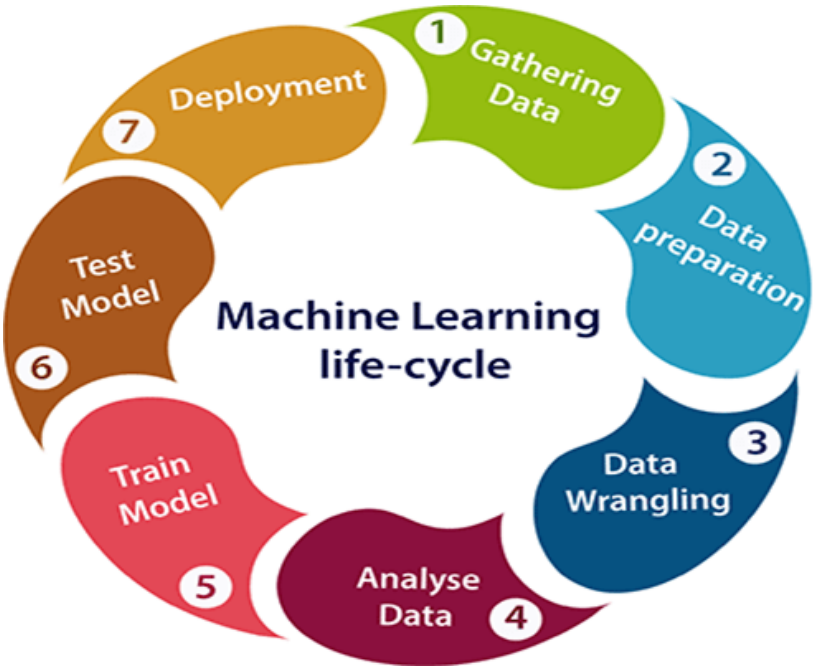
Machine learning Life cycle

Machine learning has given the computer systems the abilities to automatically learn without being explicitly programmed. But how does a machine learning system work? So, it can be described using the life cycle of machine learning. Machine learning life cycle is a cyclic process to build an efficient machine learning project. The main purpose of the life cycle is to find a solution to the problem or project.

Machine learning life cycle involves seven major steps, which are given below:

- **Gathering Data**
- **Data preparation**
- **Data Wrangling**
- **Analyse Data**

- **Train the model**
- **Test the model**
- **Deployment**



The most important thing in the complete process is to understand the problem and to know the purpose of the problem. Therefore, before starting the life cycle, we need to understand the problem because the good result depends on the better understanding of the problem.

In the complete life cycle process, to solve a problem, we create a machine learning system called "model", and this model is created by providing "training". But to train a model, we need data, hence, life cycle starts by collecting data.

1. Gathering Data:

Data Gathering is the first step of the machine learning life cycle. The goal of this step is to identify and obtain all data-related problems.

In this step, we need to identify the different data sources, as data can be collected from various sources such as **files**, **database**, **internet**, or **mobile devices**. It is one of the most important steps of the life cycle. The quantity and quality of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.

This step includes the below tasks:

- Identify various data sources
- Collect data

- Integrate the data obtained from different sources

By performing the above task, we get a coherent set of data, also called as a **dataset**. It will be used in further steps.

2. Data preparation

After collecting the data, we need to prepare it for further steps. Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.

In this step, first, we put all data together, and then randomize the ordering of data.

This step can be further divided into two processes:

- **Data exploration:**
  It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data.
  A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.

- **Data pre-processing:**
  Now the next step is preprocessing of data for its analysis.

3. Data Wrangling

Data wrangling is the process of cleaning and converting raw data into a useable format. It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step. It is one of the most important steps of the complete process. Cleaning of data is required to address the quality issues.

It is not necessary that data we have collected is always of our use as some of the data may not be useful. In real-world applications, collected data may have various issues, including:

- Missing Values
- Duplicate data
- Invalid data
- Noise

So, we use various filtering techniques to clean the data.

It is mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.

4. Data Analysis

Now the cleaned and prepared data is passed on to the analysis step. This step involves:

- Selection of analytical techniques
- Building models
- Review the result

The aim of this step is to build a machine learning model to analyze the data using various analytical techniques and review the outcome. It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification**, **Regression**, **Cluster analysis**, **Association**, etc. then build the model using prepared data, and evaluate the model.

Hence, in this step, we take the data and use machine learning algorithms to build the model.

5. Train Model

Now the next step is to train the model, in this step we train our model to improve its performance for better outcome of the problem.

We use datasets to train the model using various machine learning algorithms. Training a model is required so that it can understand the various patterns, rules, and, features.

6. Test Model

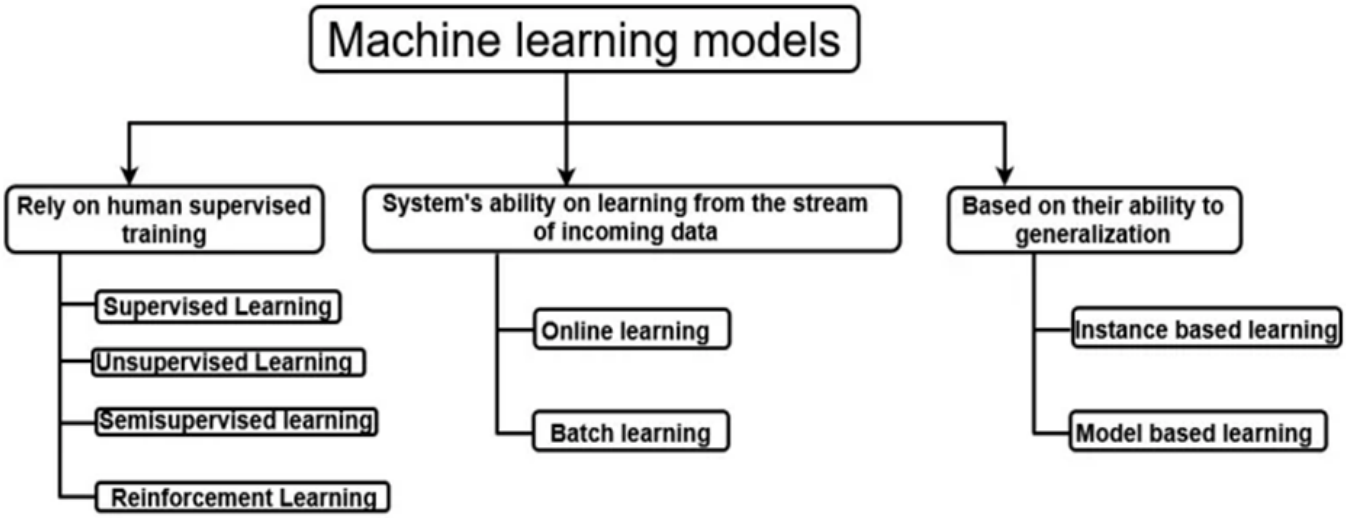
Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.

Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

7. Deployment

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.

If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.

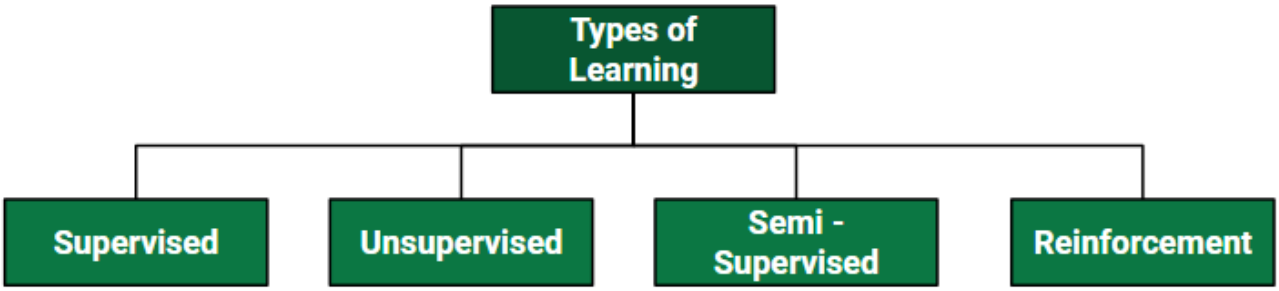


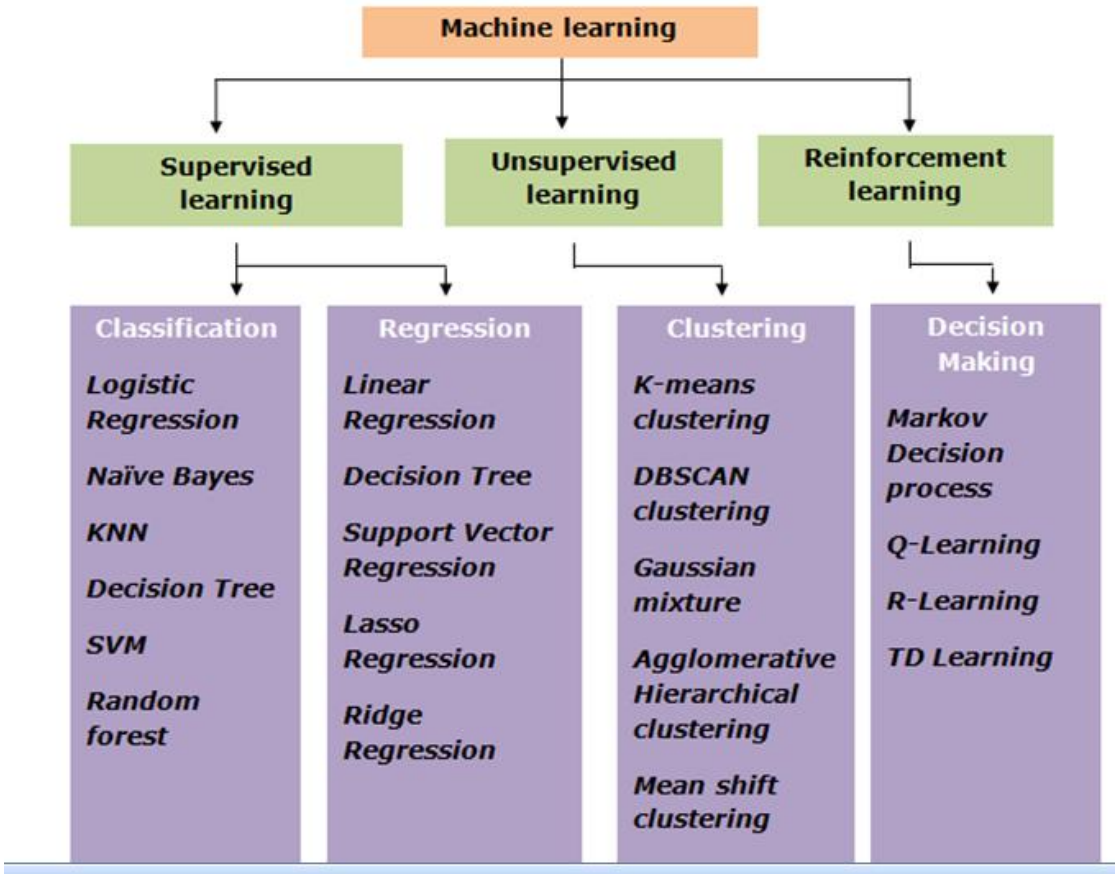
Types of Machine Learning

There are several types of machine learning, each with special characteristics and applications. Some of the main types of machine learning algorithms are as follows:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Reinforcement Learning

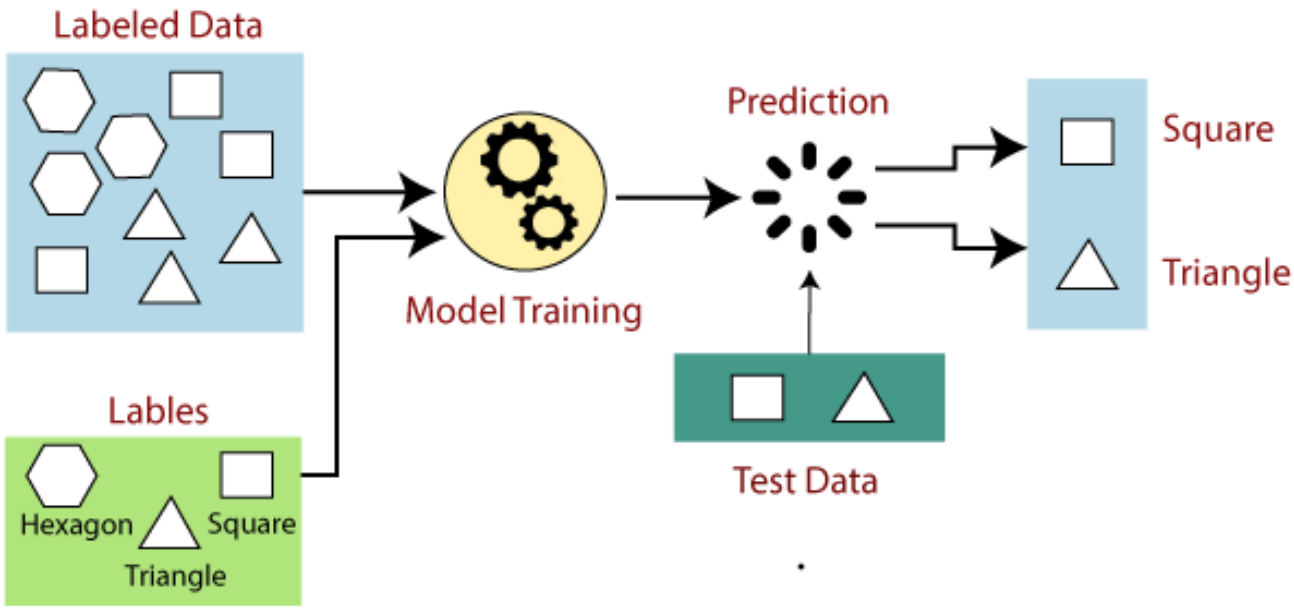
Additionally, there is a more specific category called semi-supervised learning, which combines elements of both supervised and unsupervised learning.





1. Supervised Machine Learning

As its name suggests, Supervised machine learning is based on supervision. It means in the supervised learning technique, we train the machines using the "labelled" dataset, and based on the training, the machine predicts the output. Here, the labelled data specifies that some of the inputs are already mapped to the output. More preciously, we can say; first, we train the machine with the input and corresponding output, and then we ask the machine to predict the output using the test dataset.



The main goal of the supervised learning technique is to map the input variable(x) with the output variable(y).

Categories of Supervised Machine Learning

Supervised machine learning can be classified into two types of problems, which are given below:

- **Classification**
- **Regression**

a) Classification

Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as "**Yes" or No, Male or Female, Red or Blue, etc**. The classification algorithms predict the categories present in the dataset. Some real-world examples of classification algorithms are **Spam Detection, Email filtering, etc.**

Some popular classification algorithms are given below:

- Random Forest Algorithm
- Decision Tree Algorithm
- Logistic Regression Algorithm
- Support Vector Machine Algorithm

b) Regression

Regression algorithms are used to solve regression problems in which there is a linear relationship between input and output variables. These are used

to predict continuous output variables, such as market trends, weather prediction, etc.

Some popular Regression algorithms are given below:

- Simple Linear Regression Algorithm
- Multivariate Regression Algorithm
- Decision Tree Algorithm
- Lasso Regression

Advantages and Disadvantages of Supervised Learning

Advantages:

- Since supervised learning work with the labelled dataset so we can have an exact idea about the classes of objects.
- These algorithms are helpful in predicting the output on the basis of prior experience.

Disadvantages:

- These algorithms are not able to solve complex tasks.
- It may predict the wrong output if the test data is different from the training data.
- It requires lots of computational time to train the algorithm.

2. Unsupervised Machine Learning

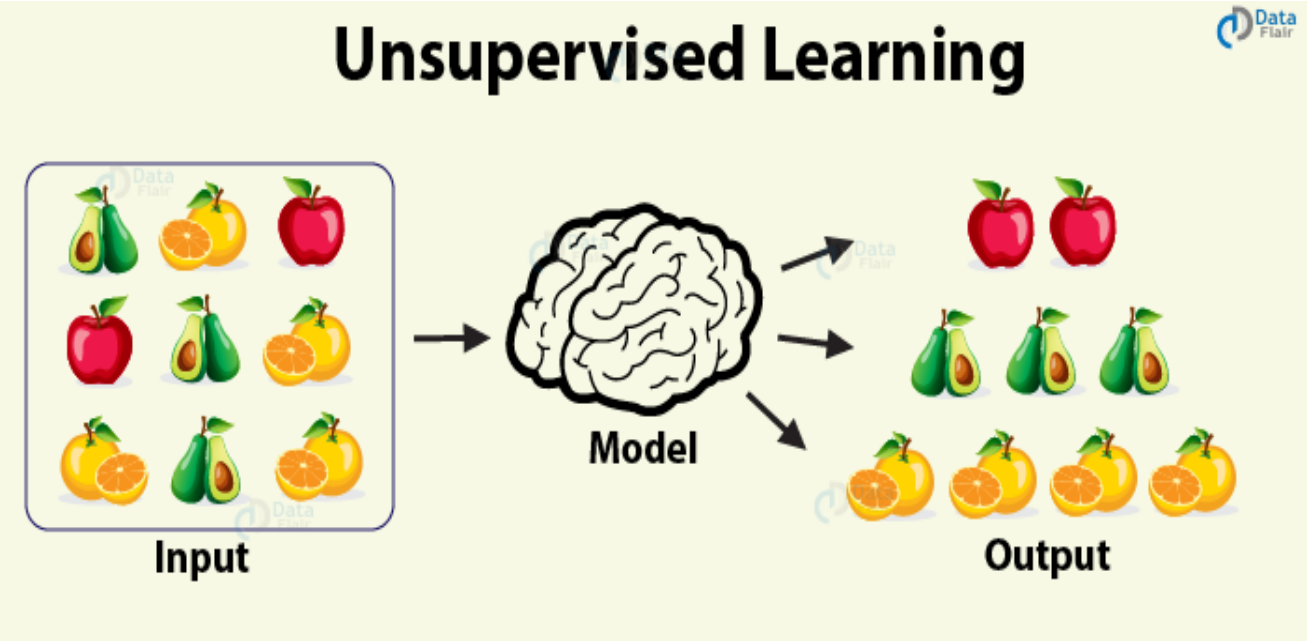
Unsupervised learning is different from the Supervised learning technique; as its name suggests, there is no need for supervision. It means, in unsupervised machine learning, the machine is trained using the unlabeled dataset, and the machine predicts the output without any supervision.

In unsupervised learning, the models are trained with the data that is neither classified nor labelled, and the model acts on that data without any supervision.

The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences. Machines are instructed to find the hidden patterns from the input dataset.

Let's take an example to understand it more preciously; suppose there is a basket of fruit images, and we input it into the machine learning model. The images are totally unknown to the model, and the task of the machine is to find the patterns and categories of the objects.

So, now the machine will discover its patterns and differences, such as colour difference, shape difference, and predict the output when it is tested with the test dataset.



Categories of Unsupervised Machine Learning

Unsupervised Learning can be further classified into two types, which are given below:

- **Clustering**
- **Association**

1) Clustering

The clustering technique is used when we want to find the inherent groups from the data. It is a way to group the objects into a cluster such that the objects with the most similarities remain in one group and have fewer or no similarities with the objects of other groups. An example of the clustering algorithm is grouping the customers by their purchasing behaviour.

Some of the popular clustering algorithms are given below:

- K-Means Clustering algorithm
- Mean-shift algorithm
- DBSCAN Algorithm
- Principal Component Analysis
- Independent Component Analysis

2) Association

Association rule learning is an unsupervised learning technique, which finds interesting relations among variables within a large dataset. The main aim of this learning algorithm is to find the dependency of one data item on another data item and map those variables accordingly so that it can generate maximum profit. This algorithm is mainly applied in **Market Basket analysis, Web usage mining, continuous production**, etc.

Some popular algorithms of Association rule learning are **Apriori Algorithm, Eclat, FP-growth algorithm.**

Advantages and Disadvantages of Unsupervised Learning Algorithm

Advantages:

- These algorithms can be used for complicated tasks compared to the supervised ones because these algorithms work on the unlabeled dataset.
- Unsupervised algorithms are preferable for various tasks as getting the unlabeled dataset is easier as compared to the labelled dataset.

Disadvantages:

- The output of an unsupervised algorithm can be less accurate as the dataset is not labelled, and algorithms are not trained with the exact output in prior.
- Working with Unsupervised learning is more difficult as it works with the unlabelled dataset that does not map with the output.

3. Semi-Supervised Learning

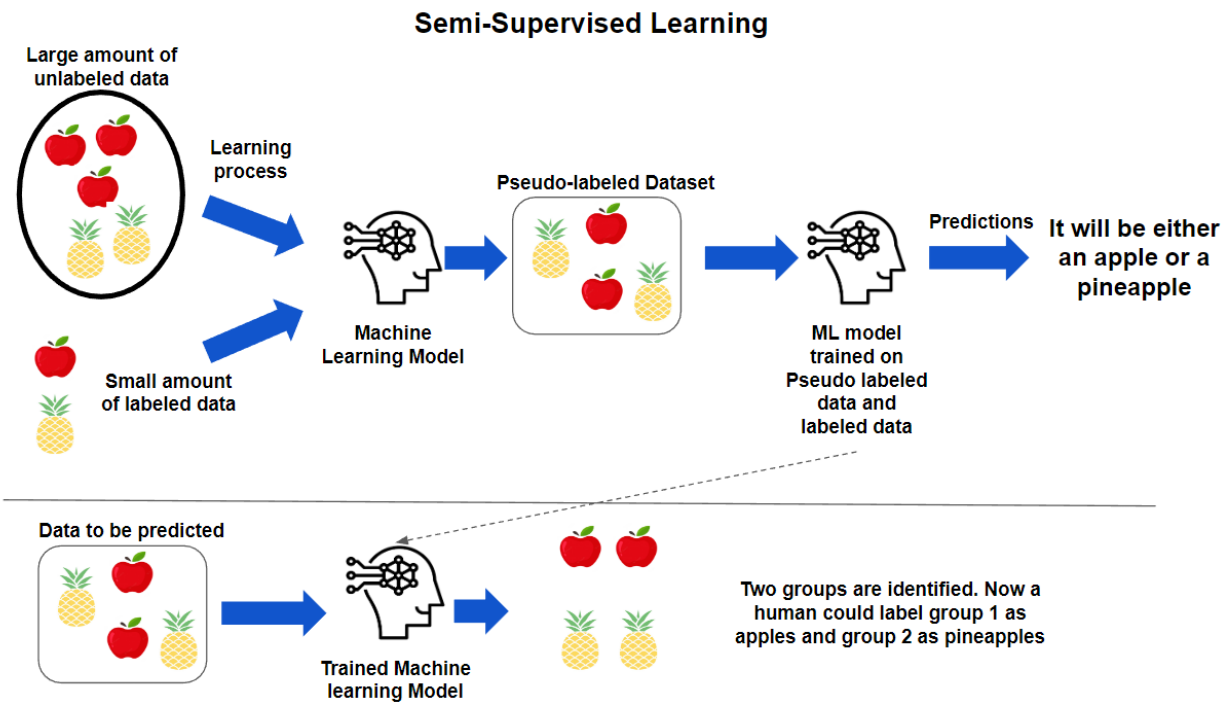
Semi-supervised learning is a Machine Learning paradigm that trains models using a small amount of labeled data together with a large amount of unlabeled data.

Key idea:
Use unlabeled data to improve learning when labeled data is expensive, time-consuming, or scarce.

Semi-Supervised learning is a type of Machine Learning algorithm that lies between Supervised and Unsupervised machine learning. It represents the intermediate ground between Supervised (With Labelled training data) and Unsupervised learning (with no labelled training data) algorithms and uses the combination of labelled and unlabeled datasets during the training period.

To overcome the drawbacks of supervised learning and unsupervised learning algorithms, the concept of Semi-supervised learning is introduced. The main aim of semi-supervised learning is to effectively use all the available data, rather than only labelled data like in supervised learning. Initially, similar data is clustered along with an unsupervised learning algorithm, and further, it helps to label the unlabeled data into labelled data. It is because labelled data is a comparatively more expensive acquisition than unlabeled data.

1. Train an initial model using labeled data
2. Predict labels for unlabeled data
3. Select confident predictions
4. Retrain model using both labeled + pseudo-labeled data



Advantages and disadvantages of Semi-supervised Learning

Advantages:

- It is simple and easy to understand the algorithm.
- It is highly efficient.
- It is used to solve drawbacks of Supervised and Unsupervised Learning algorithms.

Disadvantages:

- Iterations results may not be stable.
- We cannot apply these algorithms to network-level data.
- Accuracy is low.

4.Self-Supervised Learning

Self-Supervised Learning is a learning paradigm where a model learns from unlabeled data by creating its own supervision (labels) from the data itself.

Core idea:
No human-annotated labels are used, but the model still learns using a well-defined loss function.

What does it mean (in simple terms)?

- You don't label data manually
- The model constructs a task from the data
- It learns useful representations (features)
- These representations are later used for real tasks (classification, detection, etc.)

Two-Stage Process

1.Pretext Task (Self-created task)

Learn patterns from data itself

2.Downstream Task

Use learned features for a real ML task

Applications of Self-Supervised Learning

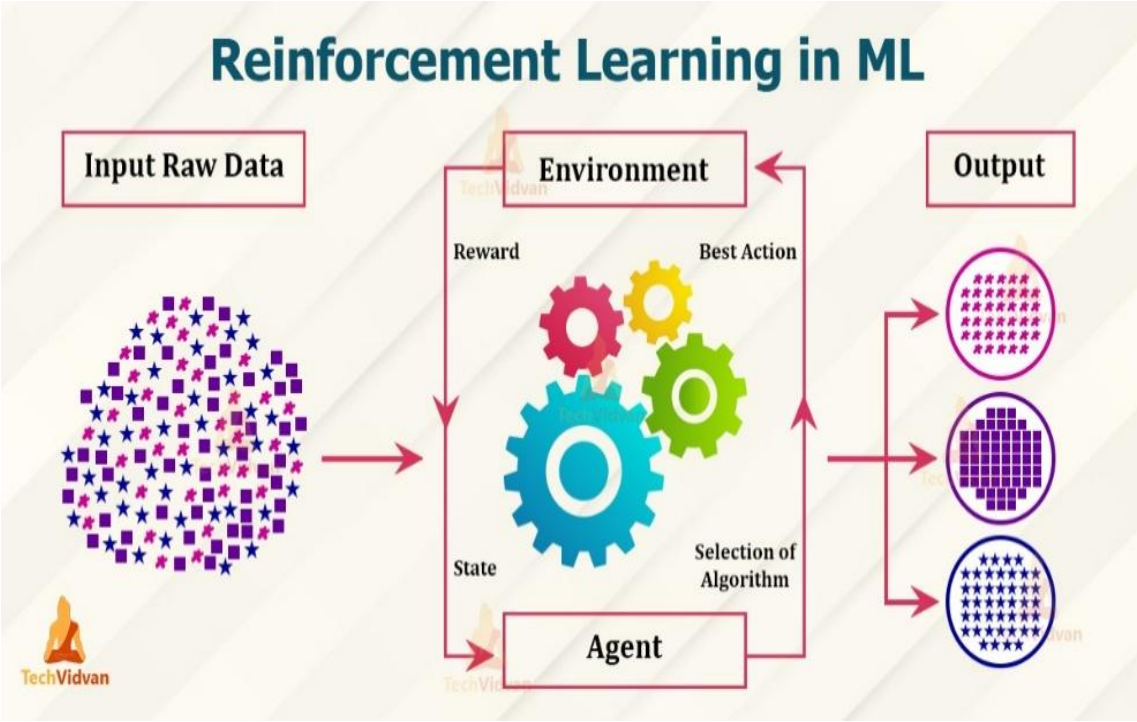
1. **Natural Language Processing (NLP)**: Pretrained models like BERT (Bidirectional Encoder Representations from Transformers), GPT, and RoBERTa rely heavily on self-supervised learning.
2. **Computer Vision**:
3. **Speech Processing**: Self-supervised methods are used for automatic speech recognition (ASR) and speech synthesis.
4. **Recommendation Systems**: Learning user-item interactions without explicit feedback.

5. **Robotics**: Self-supervised learning enables robots to understand environments by predicting outcomes of actions without explicit labels.

Benefits of Self-Supervised Learning

1. **Scalability**: Leverages large-scale unlabeled data, which is cheaper and more abundant.
2. **Improved Feature Representations**: The learned representations are more general and transferable across tasks.
3. **Reduced Dependency on Labels**: Reduces the need for expensive labeled datasets.
4. **Versatility**: Can be applied across domains like vision, language, speech and time-series data.

5.Reinforcement Learning



Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance. Agent gets rewarded for each good action and get punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards.

In reinforcement learning, there is no labelled data like supervised learning, and agents learn from their experiences only.

The reinforcement learning process is similar to a human being; for example, a child learns various things by experiences in his day-to-day life. An example of reinforcement learning is to play a game, where the Game is the environment, moves of an agent at each step define states, and the goal of the agent is to get a high score. Agent receives feedback in terms of punishment and rewards.

Due to its way of working, reinforcement learning is employed in different fields such as **Game theory, Operation Research, Information theory, multi-agent systems.**

A reinforcement learning problem can be formalized using **Markov Decision Process (MDP).** In MDP, the agent constantly interacts with the environment and performs actions; at each action, the environment responds and generates a new state.

Categories of Reinforcement Learning

Reinforcement learning is categorized mainly into two types of methods/algorithms:

- **Positive Reinforcement Learning:** Positive reinforcement learning specifies increasing the tendency that the required behaviour would occur again by adding something. It enhances the strength of the behaviour of the agent and positively impacts it.
- **Negative Reinforcement Learning:** Negative reinforcement learning works exactly opposite to the positive RL. It increases the tendency that the specific behaviour would occur again by avoiding the negative condition.

Real-world Use cases of Reinforcement Learning

- **Video Games:**
  RL algorithms are much popular in gaming applications. It is used to gain super-human performance. Some popular games that use RL algorithms are **AlphaGO** and **AlphaGO Zero**.
- **Resource Management:**
  The "Resource Management with Deep Reinforcement Learning" paper showed that how to use RL in computer to automatically learn and schedule resources to wait for different jobs in order to minimize average job slowdown.
- **Robotics:**
  RL is widely being used in Robotics applications. Robots are used in the industrial and manufacturing area, and these robots are made more powerful with reinforcement learning. There are different industries that have their vision of building intelligent robots using AI and Machine learning technology.

- **Text Mining**
  Text-mining, one of the great applications of NLP, is now being implemented with the help of Reinforcement Learning by Salesforce company.

Advantages and Disadvantages of Reinforcement Learning

Advantages

- It helps in solving complex real-world problems which are difficult to be solved by general techniques.
- The learning model of RL is similar to the learning of human beings; hence most accurate results can be found.
- Helps in achieving long term results.

Disadvantage

- RL algorithms are not preferred for simple problems.
- RL algorithms require huge data and computations.
- Too much reinforcement learning can lead to an overload of states which can weaken the results.

The curse of dimensionality limits reinforcement learning for real physical systems.

Batch and Online Learning:

Another criterion used to classify Machine Learning systems is whether or not the system can learn incrementally from a stream of incoming data.

Batch learning:

In batch learning, the system is incapable of learning incrementally: it must be trained using all the available data. This will generally take a lot of time and computing resources, so it is typically done offline. First the system is trained, and then it is launched into production and runs without learning anymore; it just applies what it has learned. This is called offline learning.

For a batch learning system to know about new data (such as a new type of spam), it needs to be trained on a new version of the system from scratch on the full dataset (not just the new data, but also the old data), then stop the old system and replace it with the new one.

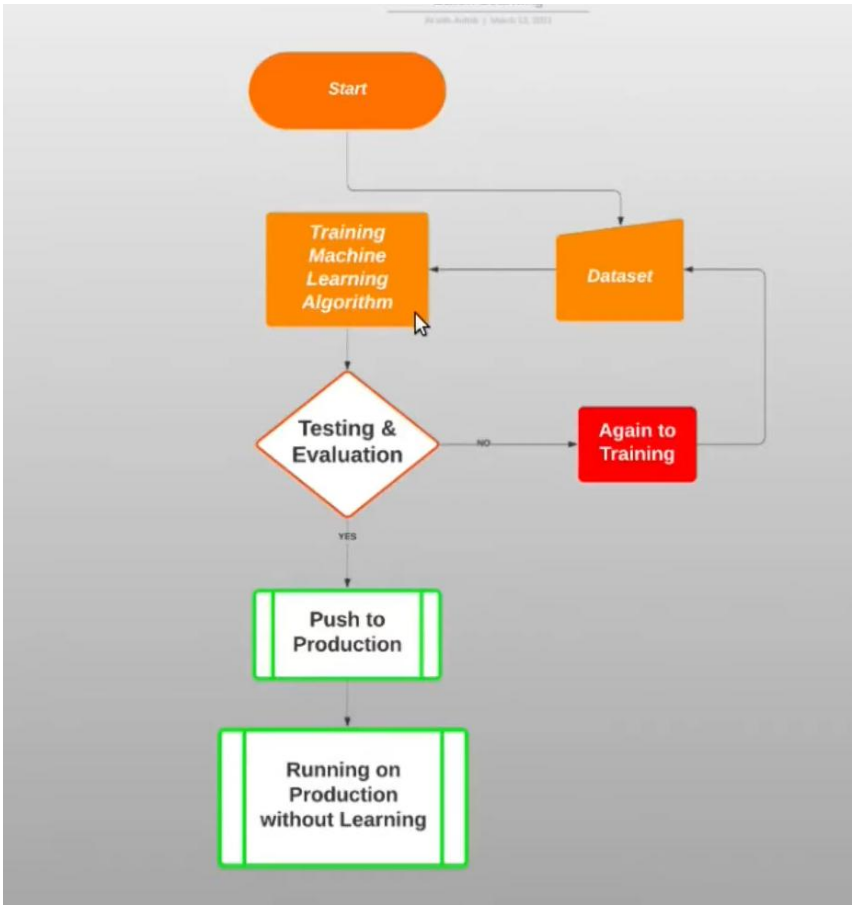
The whole process of training, evaluating, and launching a Machine Learning system can be automated fairly easily, so even a batch learning

system can adapt to change. Simply update the data and train a new version of the system from scratch as often as needed.

This solution is simple and often works fine, but training using the full set of data can take many hours, so typically train a new system only every 24 hours or even just weekly. If the system needs to adapt to rapidly changing data (e.g., to predict stock prices), then a more reactive solution is needed.

Also, training on the full set of data requires a lot of computing resources like CPU, memory space, disk space, disk I/O, network I/O, etc. If there is a lot of data and there is a need to automate the system to train from scratch every day, it will end up costing a lot of money. If the amount of data is huge, it may even be impossible to use a batch learning algorithm.

Finally, if the system needs to be able to learn autonomously and it has limited resources (e.g., a smartphone application or a rover on Mars), then carrying around large amounts of training data and taking up a lot of resources to train for hours every day is a showstopper. A better option in all these cases is to use algorithms that are capable of learning incrementally.



Program for Batch Learning

```python
from sklearn.linear_model import LinearRegression
import numpy as np

# Sample dataset: [Square Footage, Number of Bedrooms] -> House Price
X = np.array([[1400, 3], [1600, 3], [1700, 3], [1875, 4], [1100, 2], [1550, 3]])
y = np.array([245000, 312000, 279000, 308000, 199000, 219000])  # Prices in USD

# Create and train the model using batch learning (entire dataset at once)
model = LinearRegression()
model.fit(X, y)

# Make predictions
new_house = np.array([[1500, 3]])  # A house with 1500 sqft and 3 bedrooms
predicted_price = model.predict(new_house)

print(f"Predicted House Price: ${predicted_price[0]:,.2f}")
```

```
Predicted House Price: $254,445.40
```

Online learning:

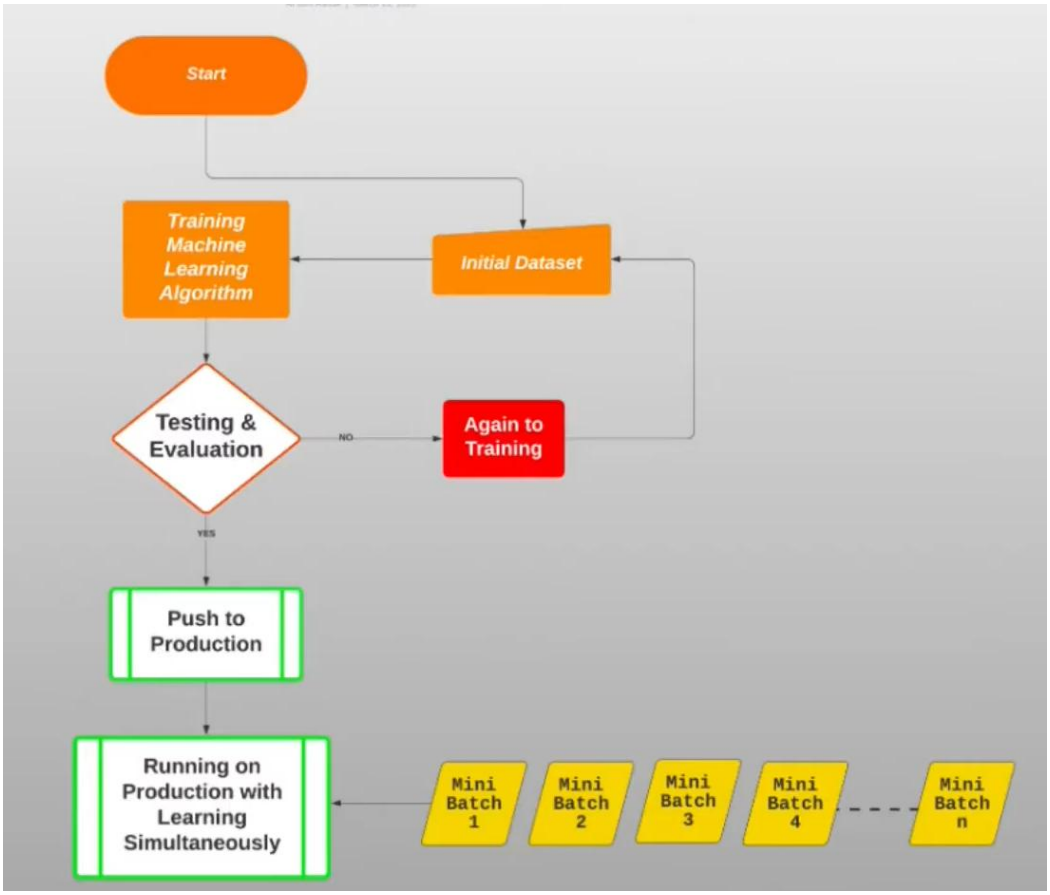
In online learning the system is trained incrementally by feeding it data instances sequentially, either individually or in small groups called mini-batches. Each learning step is fast and cheap, so the system can learn about new data on the fly, as it arrives

Online learning is great for systems that receive data as a continuous flow (e.g., stock prices) and need to adapt to change rapidly or autonomously. It is also a good option if the computing resources are limited: once an online learning system has learned about new data instances, it does not need them anymore, so they can be discarded. This can save a huge amount of space.

Online learning algorithms can also be used to train systems on huge datasets that cannot fit in one machine's main memory (this is called out-of-core learning). The algorithm loads part of the data, runs a training step on that data, and repeats the process until it has run on all of the data.

One important parameter of online learning systems is how fast they should adapt to changing data: this is called the learning rate. If learning rate is high, then the system will rapidly adapt to new data, but it will also tend to quickly forget the old data conversely, if a low learning rate is set, the system will have more inertia; that is, it will learn more slowly, but it will also be less sensitive to noise in the new data or to sequences of non-representative data points (outliers).

A big challenge with online learning is that if bad data is fed to the system, the system's performance will gradually decline. If it's a live system, the clients will notice. For example, bad data could come from a malfunctioning sensor on a robot, or from someone spamming a search engine to try to rank high in search results. To reduce this risk, the system must be monitored closely and promptly switch learning of and possibly revert to a previously working state if a drop in performance is detected. The input data can also be monitored and react to abnormal data using an anomaly detection algorithm.



Instance-Based Versus Model-Based Learning

One more way to categorize Machine Learning systems is by how they generalize. Most Machine Learning tasks are about making predictions. This means that given a number of training examples, the system needs to be able to make good predictions for (generalize to) examples it has never seen before. Having a good performance measure on the training data is good, but insufficient; the true goal is to perform well on new instances. There are two main approaches to generalization:

1. Instance-based learning and
2. Model-based learning.

1.Instance-based learning:

Instance-based learning (also known as **memory-based learning** or **lazy learning**) involves memorizing training data in order to make predictions about future data points. This approach doesn't require any prior knowledge or assumptions about the data, which makes it easy to implement and understand. However, it can be computationally expensive since all of the training data needs to be stored in memory before making a prediction. Additionally, this approach doesn't generalize well to unseen data sets because its predictions are based on memorized examples rather than learned models.

In instance-based learning, the system learns the training data by heart. At the time of making prediction, the system uses **similarity measure** and compare the new cases with the learned data. K-nearest neighbors (KNN) is an algorithm that belongs to the instance-based learning class of algorithms. KNN is a non-parametric algorithm because it does not assume any specific form or underlying structure in the data. Instead, it relies on a measure of similarity between each pair of data points. Generally speaking, this measure is based on either Euclidean distance or cosine similarity; however, other forms of metric can be used depending on the type of data being analyzed. Once the similarity between two points is calculated, KNN looks at how many neighbors are within a certain radius around that point and uses these neighbors as examples to make its prediction. This means that instead of creating a generalizable model from all of the data, **KNN looks for similarities among individual data points** and makes predictions accordingly. The picture below demonstrates how the new instance will be predicted as triangle based on greater number of triangles in its proximity.

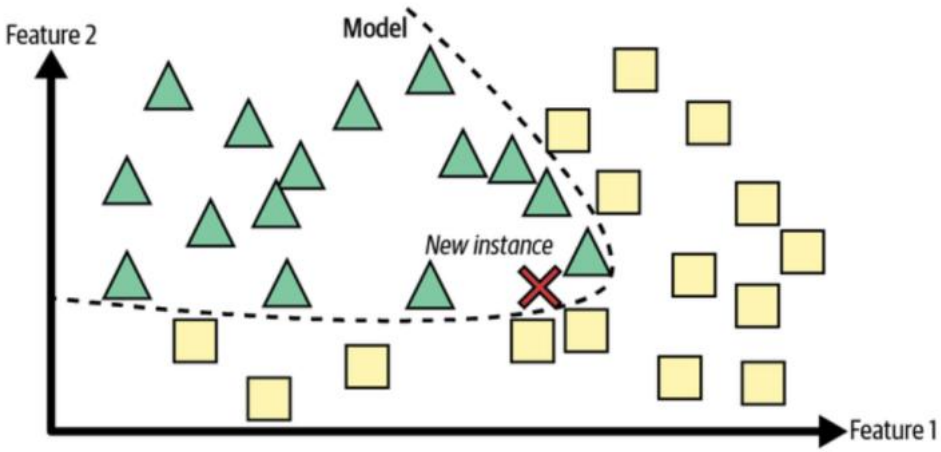
In addition to providing accurate predictions, one major advantage of using KNN over other forms of supervised learning algorithms is its versatility; KNN can be used with both numeric datasets – such as when predicting house prices – and categorical datasets – such as when predicting whether a website visitor will purchase a product or not. Furthermore, there are no parameters involved in tuning KNN since it does not assume any underlying structure in the data that needs to be fitted into; instead, all parameters involved are dependent on how close two points are considered to be similar.

Because KNN is an instance-based learning algorithm, it is not suitable for very large datasets. This is because the model has to store all of the training examples in memory, and making predictions on new data points involves comparing the new point to all of the stored training examples. However, for small or medium-sized datasets, KNN can be a very effective learning algorithm.

Other instance-based learning algorithms include **learning vector quantization (LVQ)** and **self-organizing maps (SOMs)**. These algorithms also memorize the training examples and use them to make predictions on new data, but they use different techniques to do so.

2.Model-based learning

Model-based learning (also known as structure-based or eager learning) takes a different approach by constructing models from the training data that can generalize better than instance-based methods. This involves using algorithms like linear regression, logistic regression, random forest, etc. trees to create an underlying model from which predictions can be made for new data points. The picture below represents how the prediction about the class is decided based on boundary learned from training data rather than comparing with learned data set based on similarity measures.



The model-based learning approach has several benefits over instance-based methods, such as faster processing speeds and better generalization capabilities due to its use of an underlying model rather than relying solely on memorized examples. However, this approach requires more time and effort to develop and tune the model for optimal performance on unseen data sets.

Main Challenges of Machine Learning

In **Machine Learning**, there occurs a process of analyzing data for building or training models. It is just everywhere; from Amazon product recommendations to self-driven cars, it holds great value throughout. As per the latest research, the global machine-learning market is expected to grow by 43% by 2024. This revolution has enhanced the demand for machine learning professionals to a great extent. AI and machine learning jobs have observed a significant growth rate of 75% in the past four years, and the industry is growing continuously. A career in the Machine learning domain offers job satisfaction, excellent growth, insanely high salary, but it is a complex and challenging process.

There are a lot of challenges that machine learning professionals face to inculcate ML skills and create an application from scratch. What are these challenges? we will discuss seven major challenges faced by machine learning professionals.

1. Poor Quality of Data
2. Underfitting of Training Data
3. Overfitting of Training Data
4. Machine Learning is a Complex Process
5. Lack of Training Data
6. Slow Implementation
7. Imperfections in the Algorithm When Data Grows

1. Poor Quality of Data

Data plays a significant role in the machine learning process. One of the significant issues that machine learning professionals face is the absence of good quality data. Unclean and noisy data can make the whole process extremely exhausting. We don't want our algorithm to make inaccurate or faulty predictions. Hence the quality of data is essential to enhance the output. Therefore, we need to ensure that the process of data preprocessing

which includes removing outliers, filtering missing values, and removing unwanted features, is done with the utmost level of perfection.

2. Underfitting of Training Data

This process occurs when data is unable to establish an accurate relationship between input and output variables. It simply means trying to fit in undersized jeans. It signifies the data is too simple to establish a precise relationship. To overcome this issue:

- Maximize the training time
- Enhance the complexity of the model
- Add more features to the data
- Reduce regular parameters
- Increasing the training time of model

3. Overfitting of Training Data

Overfitting refers to a machine learning model trained with a massive amount of data that negatively affect its performance. It is like trying to fit in Oversized jeans. Unfortunately, this is one of the significant issues faced by machine learning professionals. This means that the algorithm is trained with noisy and biased data, which will affect its overall performance. Let's understand this with the help of an example. Let's consider a model trained to differentiate between a cat, a rabbit, a dog, and a tiger. The training data contains 1000 cats, 1000 dogs, 1000 tigers, and 4000 Rabbits. Then there is a considerable probability that it will identify the cat as a rabbit. In this example, we had a vast amount of data, but it was biased; hence the prediction was negatively affected.

We can tackle this issue by:

- Analyzing the data with the utmost level of perfection
- Use data augmentation technique
- Remove outliers in the training set
- Select a model with lesser features

4. Machine Learning is a Complex Process

The machine learning industry is young and is continuously changing. Rapid hit and trial experiments are being carried on. The process is transforming, and hence there are high chances of error which makes the learning complex. It includes analyzing the data, removing data bias, training data, applying complex mathematical calculations, and a lot more.

Hence it is a really complicated process which is another big challenge for Machine learning professionals.

5. Lack of Training Data

The most important task you need to do in the machine learning process is to train the data to achieve an accurate output. Less amount training data will produce inaccurate or too biased predictions. Let us understand this with the help of an example. Consider a machine learning algorithm similar to training a child. One day you decided to explain to a child how to distinguish between an apple and a watermelon. You will take an apple and a watermelon and show him the difference between both based on their color, shape, and taste. In this way, soon, he will attain perfection in differentiating between the two. But on the other hand, a machine-learning algorithm needs a lot of data to distinguish. For complex problems, it may even require millions of data to be trained. Therefore, we need to ensure that Machine learning algorithms are trained with sufficient amounts of data.

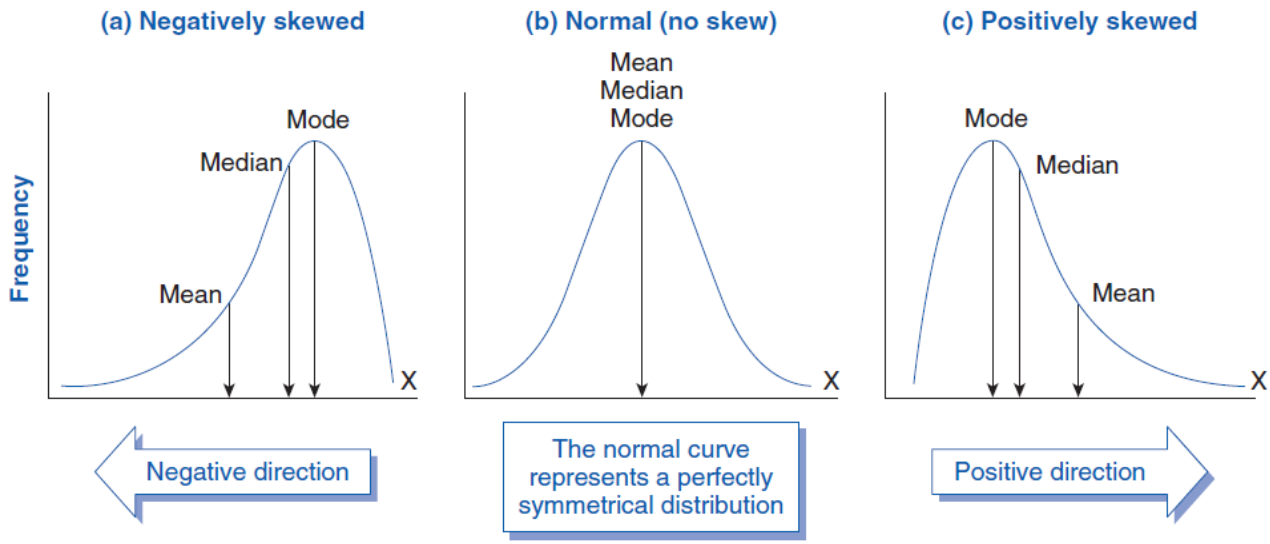
6. Slow Implementation

This is one of the common issues faced by machine learning professionals. The machine learning models are highly efficient in providing accurate results, but it takes a tremendous amount of time. Slow programs, data overload, and excessive requirements usually take a lot of time to provide accurate results. Further, it requires constant monitoring and maintenance to deliver the best output.

7. Imperfections in the Algorithm When Data Grows

So, you have found quality data, trained it amazingly, and the predictions are really concise and accurate. Yay, you have learned how to create a machine learning algorithm!! But wait, there is a twist; the model may become useless in the future as data grows. The best model of the present may become inaccurate in the coming Future and require further rearrangement. So, you need regular monitoring and maintenance to keep the algorithm working. This is one of the most exhausting issues faced by machine learning professionals.

LAB Related Topics:



4️⃣ Skewness formula

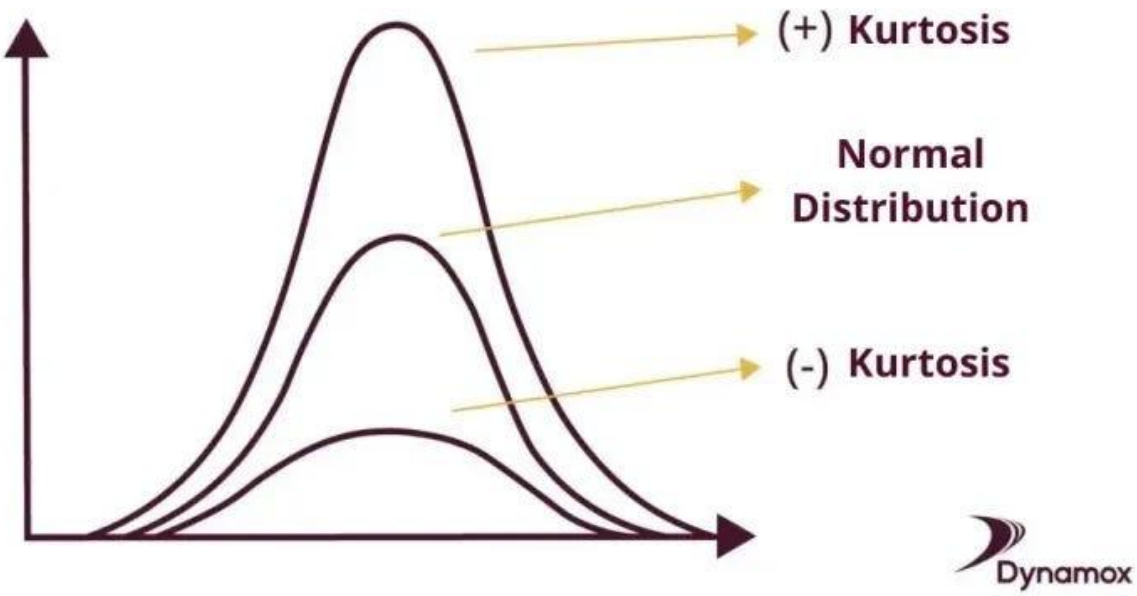
$$\text{Skewness} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^3}{\sigma^3}$$

📌 This makes skewness **unitless** and comparable across datasets.

Skewness Value	**Type**	**Meaning**
0	Symmetric	Perfectly balanced (e.g., normal distribution)
> 0	Positively skewed (Right-skewed)	Long tail to the **right**
< 0	Negatively skewed (Left-skewed)	Long tail to the **left**

4️⃣ Kurtosis (population form)

$$\text{Kurtosis} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^4}{\sigma^4}$$



Kurtosis Value	Interpretation
= 0	Normal distribution
> 0	Sharp peak, **heavy tails**, more outliers
< 0	Flat peak, **light tails**, fewer outliers